

Homework 1 — Catching a Bad Run

Fault Detection & Classification · Multivariate Statistical Process Control

ECE 8803-A14 · AI for Semiconductor Manufacturing · Fall 2026 · Prof. Asif Khan · 100 points · Due: _____

Course website: <https://aikhan123.github.io/khan-lab-website/ai-semiconductors-course.html>

A process step has finished and the tool has logged one sensor channel for the whole run: two thousand samples, from idle through ramp-up, plateau and ramp-down. Somebody has to decide whether those wafers move on, using nothing but the trace. You will build that decision, then find out what it cannot see.

The data

One file, `hw1_data.npz`. Every run is a row of 2,000 samples, already aligned in time, and the three sets are disjoint.

Array	Shape	What it is
<code>baseline_100</code>	100×2000	Healthy runs, inspected and accepted. Build your model from these.
<code>scored_100</code>	100×2000	Fifty healthy, fifty faulty. Labels in <code>scored_100_y</code> (0 healthy, 1 faulty), fault names in <code>scored_100_type</code> .
<code>blind_10</code>	10×2000	Ten runs, no labels. Six are healthy and four are not.
<code>golden</code>	2000	The nominal recipe trace, noise-free.

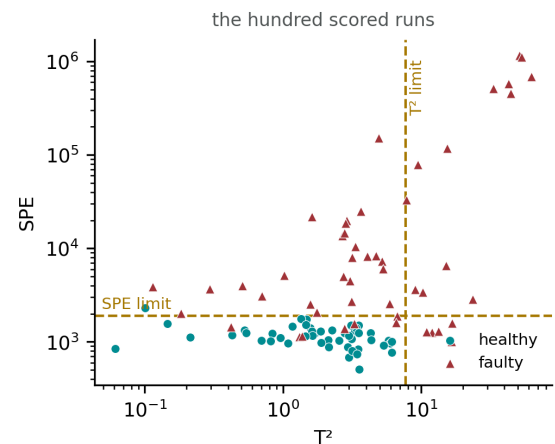
Problem 1 — Build the monitor

60 points

The starter notebook has the whole of Problem 1 already written. Run the cells in order and answer the questions marked **Report**. Read the code before you run it: Problem 2 asks you to write your own, using the same objects.

The model keeps three components of `baseline_100` and computes two numbers for every run. The first asks whether the run is an unusual *combination* of the ingredients healthy runs are made of: a normal-looking shape, present in an abnormal amount. The second rebuilds the run from the three components and adds up the square of what will not fit, which asks whether the run contains a shape the healthy population never produced. Both limits sit at the ninety-ninth percentile of the baseline scores, and a run is stopped when either number is above its limit.

- Both limits, and how much of the variation the three components carry. Read the three components against the mean run and say in a sentence each what they describe. One carries most of the variance on its own: say which part of the run it moves, and why healthy runs differ along it.
- The detection rate, the false alarm rate and the precision on `scored_100`, each as a count and a percentage, with each defined in your own words. The limits were set to stop one healthy run in a hundred; compare that against what you measured and say what could cause the difference.
- From the scatter and the per-fault-type table: name the fault type T^2 catches on its own and explain why the leftover stays small for it; say what the fault types SPE catches have in common; and work out how many of the fifty faults you would have caught with only one of the two statistics.



The scatter from item 3, which the starter notebook draws. Both axes logarithmic, dashed lines are the two limits. Upper right is out on both statistics; above one line only is out on that one alone.

Problem 2 — The blind ten

40 points

Score `blind_10` with the monitor you just built and decide which four runs to stop. For the ones you stop, keep the leftover at every sample instead of adding it up, and plot it against tick: that plot says where in the run the tool departed from what the model expected.

1. A table of ten rows: run number, both statistics as multiples of their limits, and your verdict.
2. For each run you stop, the per-sample leftover plotted against tick, and one sentence naming what you think went wrong and when.
3. Some faulty runs are caught by one statistic and passed by the other.

Identify every such run, say which statistic did the work, and explain in physical terms why the other one stayed quiet.

Worth knowing before you start. A monitor built from one of the two statistics gets some of these right. The second statistic is what tells you which ones you would otherwise have missed, and there is no way to recover that information after the fact.

Submit one notebook or one PDF, plus the code that produced it, with labelled axes on every figure. Everyone works from the same data file, so the same numbers come out for everyone. The credit is in how you explain them. Discuss the ideas freely and write your own code.

run	T ² /lim	SPE/lim	caught by	verdict
2	0.45	0.54	—	pass
5	0.65	5.84	SPE alone	STOP
...				
9				

Two rows filled in so the format is clear. Run 5 is the case item 3 is about: the leftover is nearly six times its limit while T² sits at two thirds of its own, so one statistic stops the run and the other passes it.